Determination of the Mass of Jupiter Using Observations of the Galilean Satellites

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Abstract

Jupiter is the largest planet in our solar system. It has more than 100 natural satellites. But when observed with a telescope from earth, we see mostly foour of them because others are smaller in size. These are Io, Europa, Ganymede and Callisto. These satellites revolvs around jupiter due to the gravitational attraction between the moons and the jupiter. In this project, I have estimated the mass of jupiter by observing the motion of the four satellites for 3 consecutive nights.

Contents

1 Introduction

Jupiter is the largest planet in our solar system. In fact it is so large that the center of mass of the sun-jupiter system lies outside the sun. Thus it is very important for astronomical purposes to know the mass of this planet. There were multiple attempt in history to measure the mass of the jupiter. The first recorder measurement was done by Sir Issac Newton in the late 17th century. He used the observed motions of the satellites of jupiter to estimate the mass of the planet using keplar's 3rd law. Here in this project, I have attempted to follow the same approach that newton followed, but using modern telescopes and camera. The details are discussed below.

2 Theory and Assumptions

2.1 Kepler's Third Law

The keplar's third law says for any object revolving around any giant body of mass M due to gravity has time period given by,

$$T^2 = \frac{4\pi^2 a^3}{GM} \tag{1}$$

Where 'a' is the semi major axis of the orbit and 'G' is the gravitational constant.
Now we can rearrange the formula as,

$$M = \frac{4\pi^2 a^3}{GT^2} \tag{2}$$

2.2 Assumptions

The orbits of the four satellites are highly inclined. So here I assume them as edge on (inclination $= 90^0$). I also assume that the orbit is perfectly circular. Thus when we observe the motion of each of them around the jupiter, we should see a sinusoidal behaviour.

Thus we estimate the time period of the satellites by using the formula,

$$x = A\sin(\omega t + \phi) \tag{3}$$

where the x is the distance of each satellite from the center of the jupiter in the projected sky plane. And the 'A' is the amplitude of the motion in the pixel plane (as image is taken by cmos camera).

Here also I took some observation motivated assumptions during the fitting. For example, one of them is, I restricted the frequency of the Io in the range [3, 4] day^{-1} because in the first and 3rd day we observed the transit of Io in front of the jupiter. For others one can see the actual code.

I also took the average radius of the orbits of the satellites from the literature (source: Wikipedia). They are,

Table 1: Accepted orbital radii (semi-major axes) of the Galilean satellites.

Satellite	Orbital Radius (m)
Io	4.217×10^8
Europa	6.711×10^8
Ganymede	1.0704×10^9
Callisto	1.8827×10^9

3 Observations and Data Acquisition

3.1 Observations

I did the observations with my classmates on May 11, May 12 and May 13 from 8 to 11 pm IST. The place of observation is TIFR Mumbai terrace, Mumbai, MH, India. The coordinates of the place are (18°54'22.61"N, 72°48'18.36"E) (using Google Earth)

We used the exposure time for each image in the range 0.035 sec to 0.5 sec depending on the clouds present at that time in the sky.

3.2 Instruments

The instruments we have used are listed below,

- William Optics Guidescope Guide Star Apo 61 (f/5.9, F = 360mm)
- Automated tracker mount
- A smaller guidescope to track the object.
- Two cmos camera by ZWO. One is for imaging and other is for tracking purposes.
- A laptop to control everything.

We attached the two small telescopes on top of a 14 inch Cassagrain Telescope because it was originally attached with the tracker. But we didn't used that telescope for imaging or tracking purposes. The image of the setup has shown below,

Figure 1: Instrument setup

4 Data Reduction

The observed data is reduced by using the following formula using a custom python pipeline,

$$reduced = \frac{(data - bias - dark)}{flat} \tag{4}$$

For eact measurement we took 5 images at the same time. I applied this reduction formula to all of them. And then final processed image is obtained by stacking all 5 which gave us higher signlal to noise ratio compared to indivisuals.

One of the reduced image has shoun below,

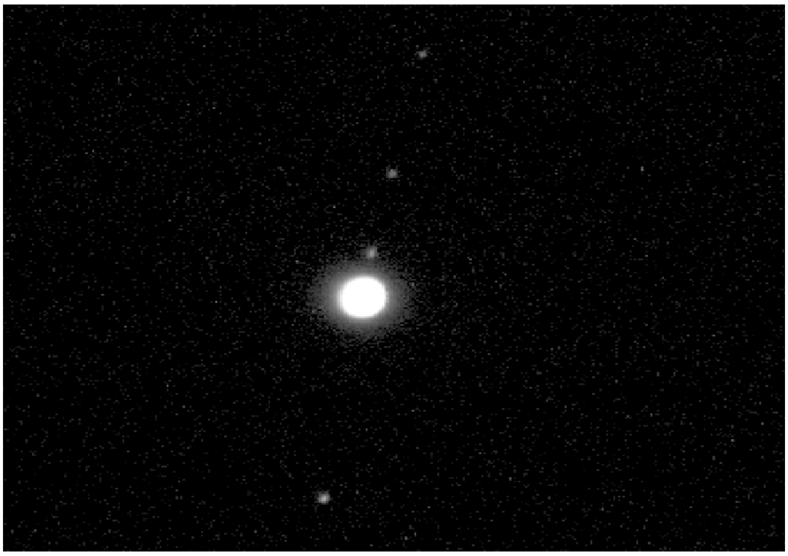
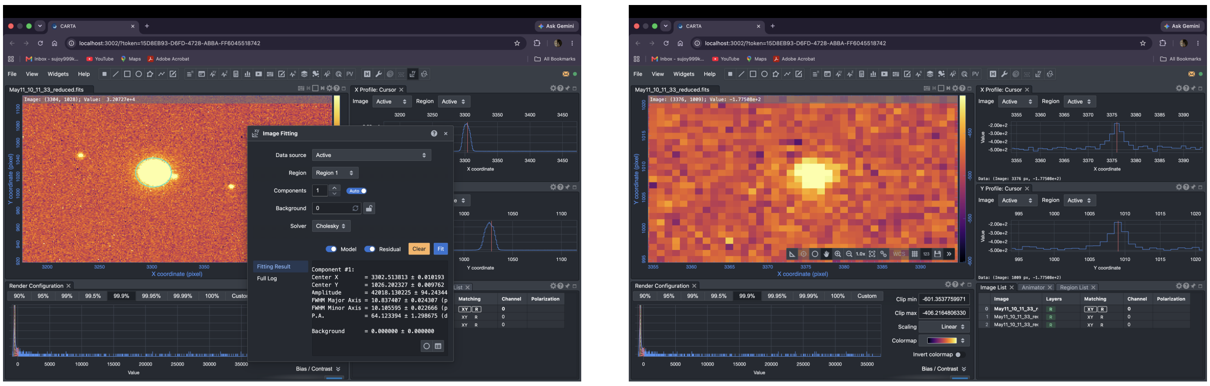


Figure 2: Processed image of Jupiter and its satellites.

5 Position Measurements

As the jupiter is too bright compared to it's satellites, so all my attempts to write python code to extract the position from the fits images failed. So instead I used CARTA to manually extract the positions of the satellites and the jupiter itself. To find the position of the jupiter, I fitted a gaussian on the image. For the satellites as they are not very bright, I used the position of the pixel with maximum brightness for them. That are shown in the images Figure 3 and Figure 4 below.



(a) Extracting the position of Jupiter.

(b) Extracting the position of satellites.

Figure 3: Position extraction procedure used for Jupiter and its satellites.

6 Orbital Motion Fitting

Here I have used the equation (3) to do the fitting on the observed positions of the satellites centered at the jupiter. The motion of the satellites wrt time is given in the table below. For the full table, look at the AstroData.xlsx in the GitHub repository.

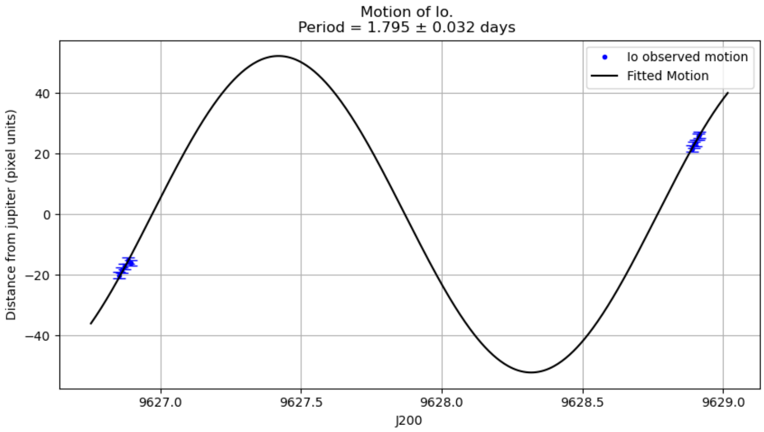
Julian Days	Io		Callistro		Ganymide		Europa	
J2000	Distance (pixel)	err (pixel)	Distance(pixel)	err (pixel)	Distance(pixel)	err (pixel)	Distance(pixel)	err (pixel)
9626.843738	-21.44	1.00	208.98	1.00	74.87	1.00	-75.78	1.00
9626.853854	-20.17	1.00	208.80	1.00	72.76	1.00	-75.47	1.00
9626.863889	-18.49	1.00	207.82	1.00	72.24	1.00	-75.85	1.00
9626.874178	-17.30	1.00	208.88	1.00	70.90	1.00	-76.13	1.00
9626.885231	-15.20	1.02	207.95	1.02	69.96	1.02	-76.20	1.02
9626.895139	-16.07	1.00	208.05	1.00	70.28	1.00	-76.47	1.00
9626.906227	NaN	NaN	207.74	1.00	67.83	1.00	-74.84	1.00
9626.915729	NaN	NaN	207.16	1.00	67.01	1.00	-75.64	1.01
9626.937836	NaN	NaN	NaN	NaN	66.99	1.01	-75.95	1.01
9627.934109	NaN	NaN	180.96	1.00	-34.42	1.00	20.00	1.00
9627.910984	NaN	NaN	181.83	1.03	-32.43	1.03	16.61	1.04
9627.907975	NaN	NaN	183.01	1.00	-32.27	1.00	16.01	1.00
9627.903391	NaN	NaN	183.64	1.00	-31.37	1.00	NaN	NaN
9627.890486	NaN	NaN	183.98	1.00	-30.36	1.00	NaN	NaN
9627.883356	NaN	NaN	184.07	1.00	-30.13	1.00	NaN	NaN
9627.875405	NaN	NaN	183.49	1.00	-29.80	1.00	NaN	NaN
9627.867905	NaN	NaN	184.29	1.00	-28.63	1.00	NaN	NaN
9627.863993	NaN	NaN	183.06	1.00	NaN	NaN	NaN	NaN
9627.859294	NaN	NaN	184.28	1.00	NaN	NaN	NaN	NaN
9628.825764	NaN	NaN	137.93	1.00	-105.74	1.00	70.67	1.00
9628.827083	NaN	NaN	137.56	1.00	-106.28	1.00	70.12	1.00
9628.833657	NaN	NaN	137.06	1.00	-105.64	1.00	70.77	1.00
9628.854526	NaN	NaN	135.93	1.00	-106.77	1.00	69.65	1.00
9628.865440	NaN	NaN	135.31	1.00	-107.40	1.00	69.03	1.00
9628.888542	21.72	1.00	133.05	1.00	-108.68	1.00	67.75	1.00
9628.896100	22.99	1.00	134.16	1.00	-110.02	1.00	67.88	1.00
9628.902847	23.61	1.00	132.80	1.00	-108.91	1.00	67.10	1.00
9628.911100	25.71	1.00	132.77	1.00	-108.94	1.00	67.65	1.00
9628.916065	26.28	1.00	131.37	1.00	-109.52	1.00	66.07	1.00

Figure 4: Position table.

7 Results

The results for each of the satellites are attached here.

7.1 Io



(a) Fit on Io data.

Io Correlation Matrix

	A	ω	φ
A	1.0	-0.42	0.42
ω	-0.42	1.0	-1.0
φ	0.42	-1.0	1.0

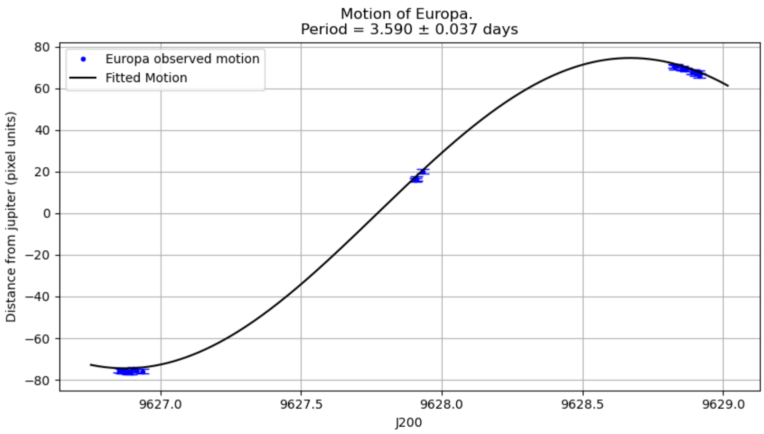
(b) Correlation matrix for Io.

Figure 5: Results of the sinusoidal fit to Io.

Parameter	Value
Period	1.795 ± 0.032 days
Estimated Mass of Jupiter	1.844e+27 ± 6.575e+25 Kg

Table 2: Results from Io.

7.2 Europa



(a) Fit on Europa data.

Europa Correlation Matrix

	A	ω	φ
A	1.0	-0.99	0.99
ω	-0.99	1.0	-1.0
φ	0.99	-1.0	1.0

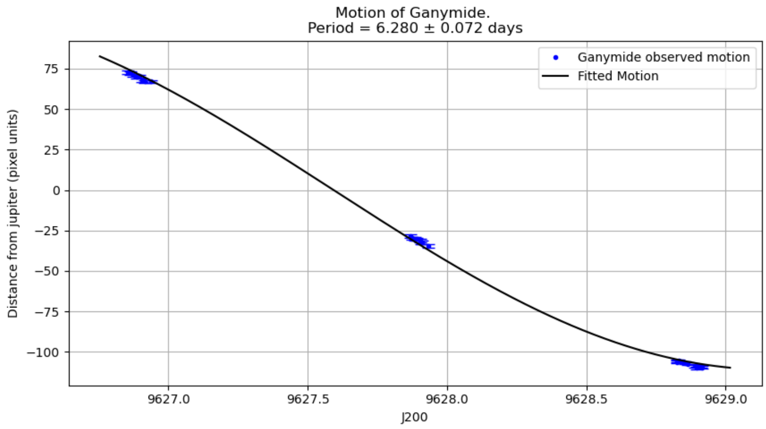
(b) Correlation matrix for Europa.

Figure 6: Results of the sinusoidal fit to Europa.

Parameter	Value
Period	3.590 ± 0.037 days
Estimated Mass of Jupiter	1.858e+27 ± 3.830e+25 Kg

Table 3: Results from Europa.

7.3 Ganymede



(a) Fit on Ganymede data.

Ganymede Correlation Matrix

	A	ω	φ
A	1.0	0.597	-0.597
ω	0.597	1.0	-1.0
φ	-0.597	-1.0	1.0

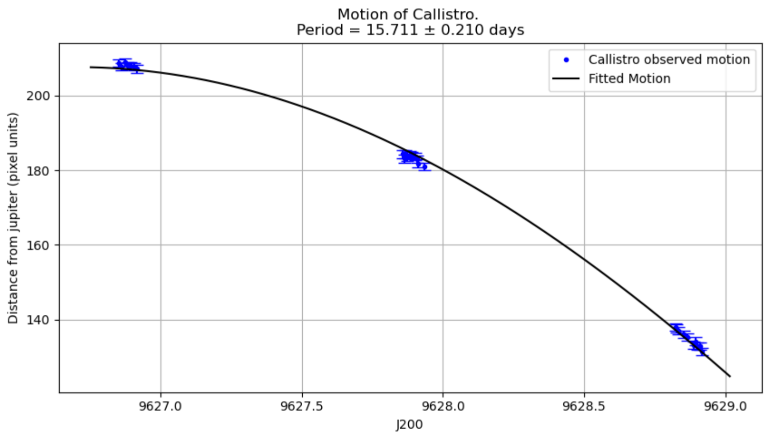
(b) Correlation matrix for Ganymede.

Figure 7: Results of the sinusoidal fit to Ganymede.

Parameter	Value
Period	6.280 ± 0.072 days
Estimated Mass of Jupiter	2.464e+27 ± 5.650e+25 Kg

Table 4: Results from Ganymede.

7.4 Callisto



(a) Fit on Callisto data.

Callisto Correlation Matrix

	A	ω	φ
A	1.0	0.748	-0.748
ω	0.748	1.0	-1.0
φ	-0.748	-1.0	1.0

(b) Correlation matrix for Callisto.

Figure 8: Results of the sinusoidal fit to Callisto.

Parameter	Value
Period	15.711 ± 0.210 days
Estimated Mass of Jupiter	2.142e+27 ± 5.727e+25 Kg

Table 5: Results from Callisto.

8 Discussion

Here we can see that the values of the orbital periods of the satellites matches quite well with literature values for the Io and Europa. Whereas for Ganymede and Callisto the deviation is large.

Table 6: Comparison of measured orbital periods with literature values.

Satellite	Observed Period (days)	Literature Period (days)	Deviation (%)
Io	1.795 ± 0.032	1.769	1.47
Europa	3.590 ± 0.037	3.551	1.10
Ganymede	6.280 ± 0.072	7.155	12.23
Callisto	15.711 ± 0.210	16.689	5.86

The main reasons for these are,

- We don't have sufficient data points to reconstruct the curve accurately.
- We observed only for 2.072327 days. The orbital period of Io is smaller than that and the orbital period of Europa is close to that. Thus for these two, the reconstruction is better.
- For the Ganymede and Callisto the orbital period is much greater than the observation period. Thus the reconstruction of the curve is not that accurate.
- Also due to insufficient data points, we can see that the correlation among the parameters for the four satelltes are quite large.

9 Final Result

Due to the above reasons mentioned in the last sections, the reliable measurements of the mass comes from only the Io and Europa. Then we use the following formula to estimate the mass of jupiter,

$$\bar{M} = \frac{\sum_i \frac{M_i}{\sigma_i^2}}{\sum_i \frac{1}{\sigma_i^2}} \tag{5}$$

$$\sigma_{\bar{M}} = \left(\sum_i \frac{1}{\sigma_i^2} \right)^{-1/2} \tag{6}$$

And the estimated mass of Jupiter is,

$$\boxed{\boldsymbol{M_J = (1.854 \pm 0.033) \times 10^{27}\ \textbf{kg}}} \tag{7}$$

Compare with accepted value:

$$M_J = 1.898 \times 10^{27}\ \text{kg} \tag{8}$$

So, the deviation from the literature value is only 2.32%.

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References

[1] Carroll, B. W. and Ostlie, D. A., *An Introduction to Modern Astrophysics.*

[2] Stellerium

[3] Celestron Sky Portal